AWS RDS Hands-on Lab Tasks

Task 1: Create an Amazon RDS Database

Launch a MySQL RDS Free Tier instance, configure credentials, networking and note the endpoint.

Task 2: Configure Security Group

Allow MySQL (3306) only from your public IP and explain why 0.0.0.0/0 is insecure.

Task 3: Connect to RDS

Connect using MySQL Workbench/DBeaver/MySQL CLI and verify using SHOW DATABASES and SELECT VERSION().

Task 4: Create Database Objects

Create a database, table, insert records, then perform SELECT, UPDATE and DELETE operations.

Task 5: Modify the RDS Instance

Change instance class, storage or backup settings and observe the modification process.

Task 6: Enable Automated Backups

Configure 7-day retention and understand automated backups.

Task 7: Create a Manual Snapshot

Create and verify a manual DB snapshot.

Task 8: Restore from Snapshot

Restore a new RDS instance from the snapshot and verify the data.

Task 9: Monitor Performance

View CPU, connections, storage and IOPS metrics in CloudWatch.

Task 10: Performance Insights

Enable Performance Insights and review top SQL and database load.

Task 11: Read Replica (Optional)

Create a Read Replica and verify replication.

Task 12: Cleanup

Delete the restored instance, retain snapshots, then remove the original instance safely.